FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2023

TEST DATE: 02-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. Potential difference across capacitor in steady state is 4V

$\therefore$ q = 8μC

2. C

Sol. Force acting on the particle at a distance 'r' from centre of core is

$$F = -\left(\frac{G\times\frac{4}{3}\pi r^3\times 12\rho\times m}{r^2}\right)$$

$a = -16\pi G\rho r$

$\omega = \sqrt{16\pi\rho G} = 4\sqrt{\pi\rho G}$

Time taken to move from A to B is

$$t = \frac{\pi}{\omega} = \frac{1}{4}\sqrt{\frac{\pi}{\rho G}}$$

3. A

Sol. Loss in potential energy = Gain in kinetic energy

$$\frac{q^2}{3\pi\varepsilon_0\ell} = \frac{1}{2}\left\{\frac{m\ell^2}{4}+\frac{m\ell^2}{4}\right\}\omega^2 = \frac{m\ell^2\omega^2}{4}$$

$$\omega = \sqrt{\frac{4q^2}{3\pi\varepsilon_0 m\ell^3}}$$

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4. C

Sol. From conservation of momentum

$mv_0 = (M+m)v$

We know,

$v^2 = \omega^2(A^2 - y^2)$

$$x\left(\frac{mv_0}{M+m}\right)^2 = \frac{k}{M+m}\left(A^2 - \frac{m^2g^2}{k^2}\right)$$

Hence $y = \frac{mg}{k}$

$$\therefore A = \frac{m}{k}\sqrt{\frac{v_0^2 k}{M+m} + g^2}$$

5. D

Sol. The possible frequencies of other three forks are 553 Hz, 555 Hz and 560 Hz.

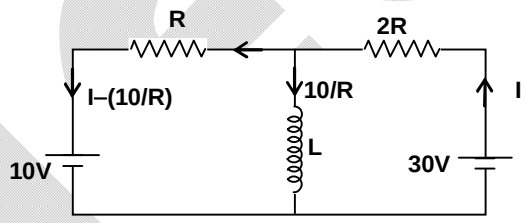
6. A

Sol. $30 - I(2R) - R\left(I - \frac{10}{R}\right) - 10 = 0$

$$\Rightarrow \quad I = \frac{10}{R}$$

Voltage across inductor = 10 V

$$\frac{dI}{dt} = 10 \text{ A/s}$$



7. C

Sol. $V_L^2 + V_R^2 = V_1^2 = (150)^2$ …(i)

$V_C^2 + V_R^2 = V_2^2 = (130)^2$ …(ii)

$V_3 = V_R = 120$ …(iii)

Solving (i), (ii) and (iii) we get

$V_L = 90V, V_C = 50\,V$

$$\therefore \quad V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(120)^2 + (40)^2}$$

$\Rightarrow V = 40\sqrt{10}$ V

$$\cos\phi = \frac{120}{40\sqrt{10}} = \frac{3}{\sqrt{10}}$$

8. B

Sol. $\nu_1 = \frac{v}{4\ell}$, $\nu_2 = \frac{v}{4(\ell + x)}$

$\therefore$ Beat frequency

$\beta = \nu_1 - \nu_2$

$$= \frac{v}{4\ell} - \frac{v}{4(\ell+x)} = \frac{vx}{4\ell^2} \text{ (Since, } x << \ell)$$

9. D

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10. D

Sol. $\frac{E}{B} = C \Rightarrow E = BC = 6$

$\therefore \vec{E} = -6\sin\left[200\pi(y + ct)\right]\hat{k}$ V/m

11. D

Sol. $y = y_0 \sin^2 \omega t = \frac{y_0}{2}(1 - \cos 2\omega t)$

Mean position is shifted by $\frac{y_0}{2}$ and frequency is 2ω

At equilibrium, $\frac{ky_0}{2} = mg \Rightarrow \frac{k}{m} = \frac{2g}{y_0}$

Also, spring constant

$k = m(2\omega)^2$

$\Rightarrow 2\omega = \sqrt{\frac{2g}{y_0}}$

$\therefore$ time period is $T = \frac{2\pi}{2\omega} = 2\pi\sqrt{\frac{y_0}{2g}}$

12. D

Sol. $m_1 + m_2 = \frac{m_1\sigma}{\sigma_1} + \frac{xm_2\sigma}{\sigma_2}$

Here x = degree of submergence

$\frac{m_1}{m_2} + 1 = \frac{m_1}{m_2}\left(\frac{\sigma}{\sigma_1}\right) + x\left(\frac{\sigma}{\sigma_2}\right)$

$\Rightarrow \frac{1}{2} + 1 = \frac{1}{2}(2) + x(0.5)$

$\therefore x = 1$

13. D

Sol. Tension at mid point is

$T = \frac{3m\omega^2\ell}{8}$

$\therefore$ strain $= = \frac{3m\omega^2\ell}{8AY}$

14. C

Sol. $\left(P_0 + \frac{4\sigma}{R}\right)\left(\frac{4}{3}\pi R^3\right) = \left(\frac{P_0}{2} + \frac{4\sigma}{5R/4}\right)\frac{4}{3}\pi\left(\frac{5R}{4}\right)^3$

Solving, we get, $P_0 = \frac{96\sigma}{R}$

15. B

Sol. No. of moles is conserved,

$\frac{P_0V_0}{T_0} + \frac{2P_0V_0}{T_0} = \frac{P_0(2V_0)}{T}$

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$\frac{P}{T}=\frac{3P_0}{2T_0}$ …(i)

Internal energy remains same

$(1)\left(\frac{3R}{2}\right)(T_0)+(2)\left(\frac{5R}{2}\right)\left(\frac{T_0}{2}\right)=(3)\left(\frac{13R}{6}\right)(T)$

$\Rightarrow T=\frac{8T_0}{13}$

From (i), $P=\frac{12P_0}{13}$

16. D

Sol. Given circuit represents NOR gate.

17. A

Sol. $A_{max}=A_C+A_m=300+200=500V$

$A_{min}=A_C-A_m=300-200=100V$

$\therefore \frac{A_{max}}{A_{min}}=5$

18. A

Sol. For 1st maxima,

$a\sin\theta=\frac{3\lambda}{2}$

For 1st case,

$a\sin\theta_1=\frac{3\lambda_1}{2}$

$\Rightarrow \frac{ay_1}{D}=\frac{3\lambda_1}{2}$

$y_1=\frac{3\lambda_1 D}{2a}$

Similarly, $y_1=\frac{3\lambda_2 D}{2a}$

$\therefore \Delta y=y_2-y_1=\frac{3D}{2a}(\lambda_2-\lambda_1)=1.5\times10^{-5}$ m

19. C

Sol. $\tan\theta=\frac{dy}{dx}=x$

$a_t=g\sin\theta=\frac{gx}{\sqrt{x^2+1}}$

20. A

Sol. $L=\tau\Delta t$

$k=\frac{L^2}{2I}$

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SECTION – B

21. 5

Sol. $A = \sqrt{(3)^2 + (4)^2} = 5$ cm

$\therefore$ 100 A = 500 cm = 5 m

22. 4

Sol. $a_t = \sqrt{3}t$

Also, $a_C = \left(\frac{\sqrt{3}}{2}t^2\right)^2$

$\therefore\ t = (2)^{2/3}$

$\Rightarrow t^3 = (2)^2 = 4$

23. 60

Sol. For no toppling;

$F\left(\frac{a}{2}+b\right) \le mg\left(\frac{a}{2}\right)$

$\mu mg\left(\frac{a}{2}+b\right) \le mg\left(\frac{a}{2}\right)$

$\mu\left(\frac{a}{2}+b\right) \le \frac{a}{2}$

$0.4(a + 2b) \le a$

$\frac{b}{a} \le 0.75$

$\therefore\ \frac{80b}{a} \le 0.75 \times 80 = 60$

24. 5

Sol. For no. emergence of ray from AC;

$r_2 \ge \theta_C$, $\theta_C \Rightarrow$ critical angle

Also, $r_1 + r_2 = 75°$

$\Rightarrow r_2 = 75° - r_1$

$\therefore\ 75° - r_1 \ge 45°$

$\Rightarrow r_1 = 30°$

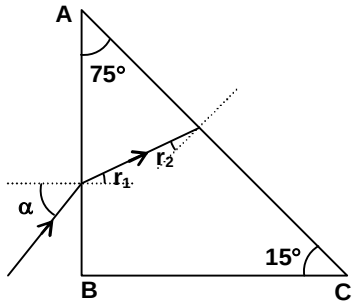
From snell's law

$\sqrt{2}\sin r_1 = \sin\alpha$

$\Rightarrow \sin\alpha \le \sqrt{2} \times \frac{1}{2}$

$\alpha \le 45°$

$\therefore\ \alpha_{max} = 45° = \frac{\pi}{4}$



25. 25

Sol. ρU = constant

$\Rightarrow \frac{T}{V}$ = constant

$\therefore$ P = constant

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$$\frac{\Delta U}{W}=\frac{nC_V\Delta T}{nR\Delta T}=\frac{C_V}{R}=\frac{5}{2}=2.5$$

$x = 2.5$

$\Rightarrow 10\,x = 25$

26. 10

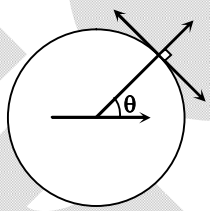
Sol. $$\frac{k_\alpha}{k_d}=\frac{m_\alpha}{m_d}$$

$$\therefore\ f=\frac{k_\alpha}{k_\alpha+k_d}=\frac{m_\alpha}{m_\alpha+m_d}=\frac{4}{200}=0.02$$

$\therefore\ 500f = 500(0.02) = 10$

27. 2

Sol. Tangential components must cancel.



$$\Rightarrow \frac{\sigma}{2\varepsilon_0}\sin\theta=\frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{R^3}$$

$$\Rightarrow R^3=\frac{p}{2\pi\sigma}$$

$\Rightarrow n = 2$

28. 18

Sol. $$\frac{dx}{x}=\frac{3(dy)}{y}+\frac{1}{2}\left(\frac{dz}{z}\right)=3\times4+\frac{1}{2}\times12=18$$

29. 100

Sol. $$n=\frac{\text{Pitch}}{\text{Least count}}=\frac{1\,\text{mm}}{10\,\mu\text{m}}=100$$

30. 6

Sol. $${}^nC_2=\frac{n(n-1)}{2}=\frac{4(4-1)}{2}=6$$

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Chemistry

PART – B

SECTION – A

31. A

Sol. $O_2^+ < O_2 < O_2^-$

32. D

Sol. Conceptual

33. B

Sol. $XeF_6 \longrightarrow XeF_5^+ + F^-$

$F^- + HF \longrightarrow HF_2^-$

34. C

Sol. pK_b of fluoride ion is = 10.83

$K_b = 10^{-10.83} = 1.48 \times 10^{-11}$

$HF + H_2O \rightleftharpoons H_3O^+ + F^-$

$K_w = K_a \times K_b$

$K_a = \frac{K_w}{K_b} = \frac{10^{-4}}{1.48 \times 10^{-11}} = 6.75 \times 10^{-4}$

35. A

Sol. Conceptual

36. D

Sol. Conceptual

37. A

Sol. pH = 14, $\left[OH^-\right] = 1$

$Ca(OH)_2 \longrightarrow Cu^{2+} + 2OH^-$

$K_{sp} : \left[Cu^{2+}\right][1]^2$

$\left[Cu^{2+}\right] : \frac{K_{sp}}{1^2} - \frac{1 \times 10^{-19}}{1} = 10^{-19}$

$Cu^{2+} + 2e^- \longrightarrow Cu$

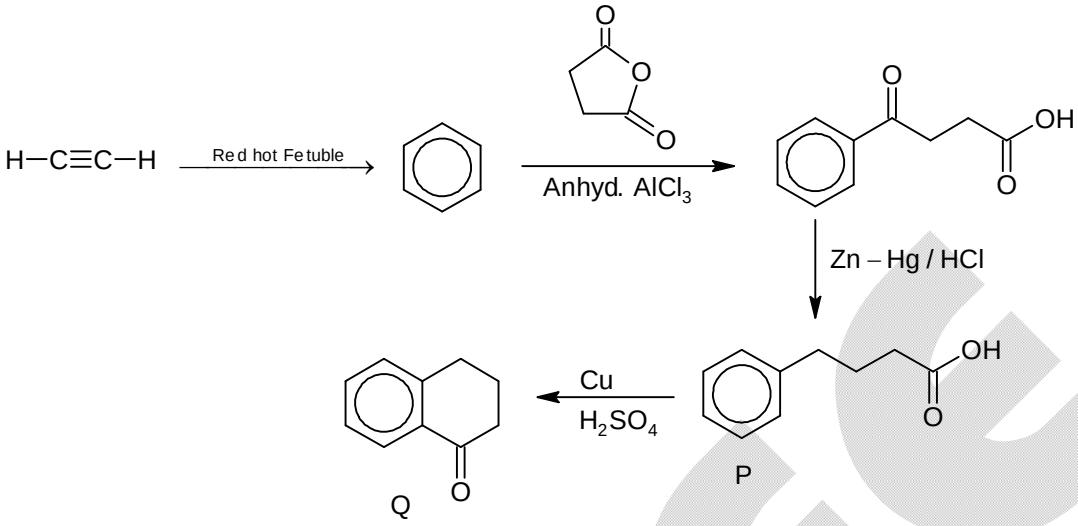
$E_{Cu^{2+}/Cu} = E^o_{Cu^{2+}/Cu} - \frac{0.0591}{2} \log \frac{1}{\left[Cu^{2+}\right]}$

$= -0.2214$ V

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38. B

Sol.

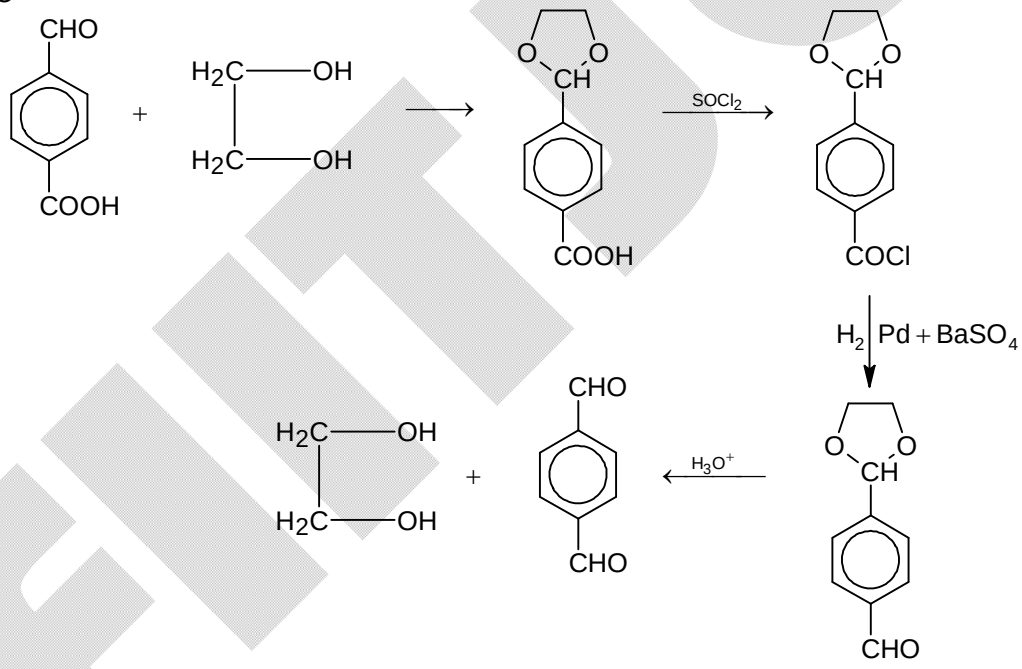


39. B

Sol. Conceptual

40. C

Sol.



41. D

Sol.

$$X(g) \longrightarrow 2Y(g) + Z(g)$$

$t = 0$ 3

$t = 10s$ $3 - a$ $2a$ a

$$\frac{r_1}{r_2} = \frac{1}{2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$$

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$$\frac{1}{2}=\frac{3-a}{2a}\sqrt{\frac{4}{16}}$$

$$\frac{1}{2}=\frac{3-a}{2a}\times\frac{1}{2}$$

$3 - a = 2a$

$a = 1$

$n_T = 3 - a + 2a + a = 5$

$$P=\frac{nRT}{V}=\frac{5\times0.08\times300}{10}=12 \text{ atm}$$

$$k=\frac{1}{t}\ell n\frac{A_0}{A}=\frac{1}{10}\ell n\frac{3}{2}=\frac{1}{10}(1.1-0.7)=\frac{0.4}{10}$$

$k = 4\times10^{-2}$ sec^{-1}

$$r=+\frac{dy}{dt}=\frac{2}{10}=0.2 \text{ mole } L^{-1} \text{ sec}^{-1}$$

42. A

Sol. $N_2(g)+3Cl_2(g)\longrightarrow 2NCl_3(g);\Delta H_F$

$$\Delta H_f \text{ of } NCl_3=\frac{1}{2}\Delta H_r=\frac{1}{2}\left[-2\times\Delta H_1+\Delta H_2-\Delta H_3\right]$$

$$\Delta H_f \text{ of } NCl_3=-\Delta H_1+\frac{\Delta H_2}{2}-\frac{\Delta H_3}{2}$$

43. D

Sol. More crystal field splitting is obeserbed when

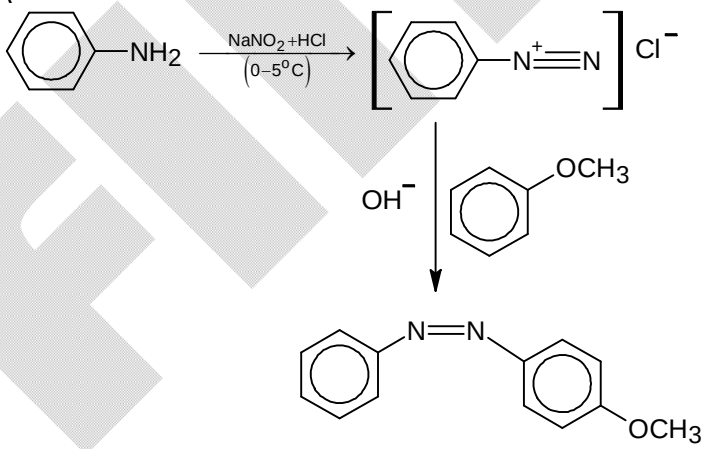
(i) More charge density on CMI/atom.

(ii) More field strength of ligands crystal field splitting increases, absorbed energy increases Wavelength decreases.

$[FeCl_6]^{4-} > [Fe(H_2O)_6]^{2+} > [Fe(H_2O)_6]^{3+}$

44. A

Sol.



45. D

Sol. $$k=\left(\frac{k_1k_2}{k_3}\right)^{2/3}$$

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$$Ae^{-E_a/RT} = \left[\frac{Ae^{-E_{a_1}/RT} \times Ae^{-E_{a_2}/RT}}{Ae^{-E_{a_3}/RT}}\right]^{2/3}$$

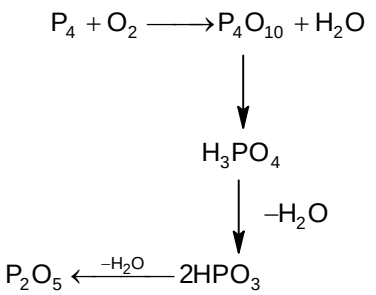
$$= \left[Ae^{\left(-E_{a_1}-E_{a_2}+E_{a_3}\right)/RT}\right]^{2/3}$$

$$E_a = \frac{2}{3}\left[E_{a_1} + E_{a_2} - E_{a_3}\right]$$

= 140

46. B

Sol.



47. A

Sol. $t_{93.75\%} = 4t_{50\%}$

$= 4 \times 120 = 480$ min

48. D

Sol. $3Na_4\left[Fe(CN)_6\right] + 4FeCl_3 \longrightarrow \underset{\text{(Prussian blue)}}{Fe_4\left[Fe(CN)_6\right]} + 12NaCl$

49. A

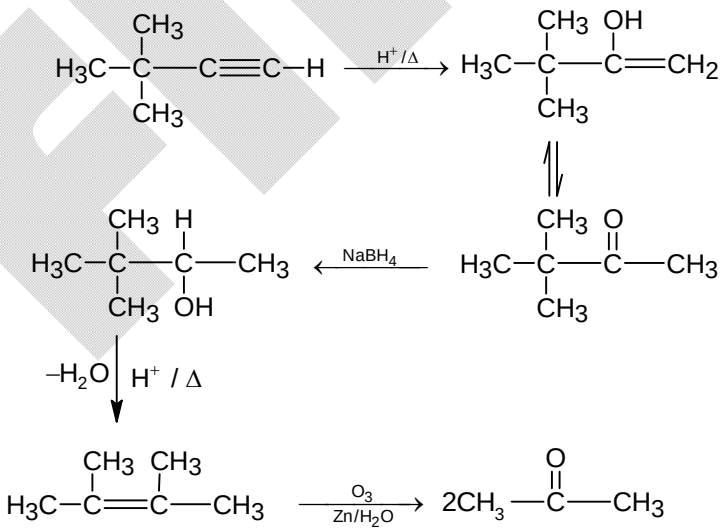
Sol. $(I.E.)_H - (E_1)_H = x$

$E \propto Z^2$

$\Delta E = E_6 - E_3$

50. D

Sol.



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SECTION – B

51. 4

Sol. $C_xH_y + O_2 \longrightarrow CO_2 + H_2O$

Applying POAC

$x \times C_xH_y = 40$

$x \times 20 = 40$

$x = 2$

$y \times C_xH_y = 2 \times 20$

$y \times 20 = 40$

$y = 2$

52. 5

Sol. de-Broglie wave length

$$\lambda = \sqrt{\frac{150}{V}}\,\text{Å} = \sqrt{\frac{150}{6}}\,\text{Å} = 5\,\text{Å}$$

$$\Delta x.\Delta P \geq \frac{h}{4\pi}$$

$$\Delta P = \frac{h}{\lambda^2}, \Delta\lambda$$

$$\therefore \Delta\lambda = 0.125 \times 5\ \text{Å}$$

53. 9

Sol. $m = \frac{x_A}{x_\beta} \times \frac{1000}{M_A}$ (M_A = Molar mass of solvent)

(M_B = Molar mass of solute)

$$m = \frac{1000}{9M_4} \quad \text{... (1)}$$

$$M = \frac{n_B}{n_A\ M_A + n_B \times M_B} \frac{1000 \times d}{}$$

$$= \frac{x_B \times 1000d}{x_A.M_A \times x_B.M_B}$$

$$M = \frac{0.1 \times 1000 \times 2}{0.9M_A + 0.1M_B}$$

$$= \frac{200}{0.9\,M_A + 0.1M_B}$$

$$= \frac{2000}{9\,M_A + M_B}$$

At m = M

$$\frac{1000}{9\,M_A} = \frac{2000}{9\,M_A + M_B}$$

$$9M_A + M_B = 18\,M_A$$

$$M_B = 9\,M_A$$

$$\frac{M_B}{M_A} = 9$$

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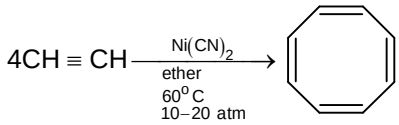
54. 27

Sol. x = 9, y = 3

xy = 27

55. 4

Sol. Cyclooctateraene is formed.



56. 14

Sol. Let α is the degree of dissociation

$$Cl-CH_2-COOH \rightleftharpoons CH_2Cl-COO^- + H^+$$

$$C(1-\alpha) \qquad C\alpha \qquad C\alpha$$

$$K_a = \frac{C\alpha.C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{1-\alpha}$$

We know that

$$\alpha = \frac{\pi^C}{\lambda_\infty} \therefore K_a = \frac{C\left(\frac{\lambda_C}{\lambda_\infty}\right)}{1-C\left(\frac{\lambda_C}{\lambda_\infty}\right)}$$

$$\frac{K_a}{C} = \frac{\lambda_{C^2}}{\lambda_\infty(\lambda_\infty-\lambda_C)}$$

$mx = y$

$K_a \; m = 0.00014 = 14\times10^{-5}$

57. 3

Sol. The edge length of original cube $= 1m = 10^3$ mm

Total surface are $= 6\times(10^3)^2 = 6\times10^6$ mm^2

When cube is broken into 10^9 cubes

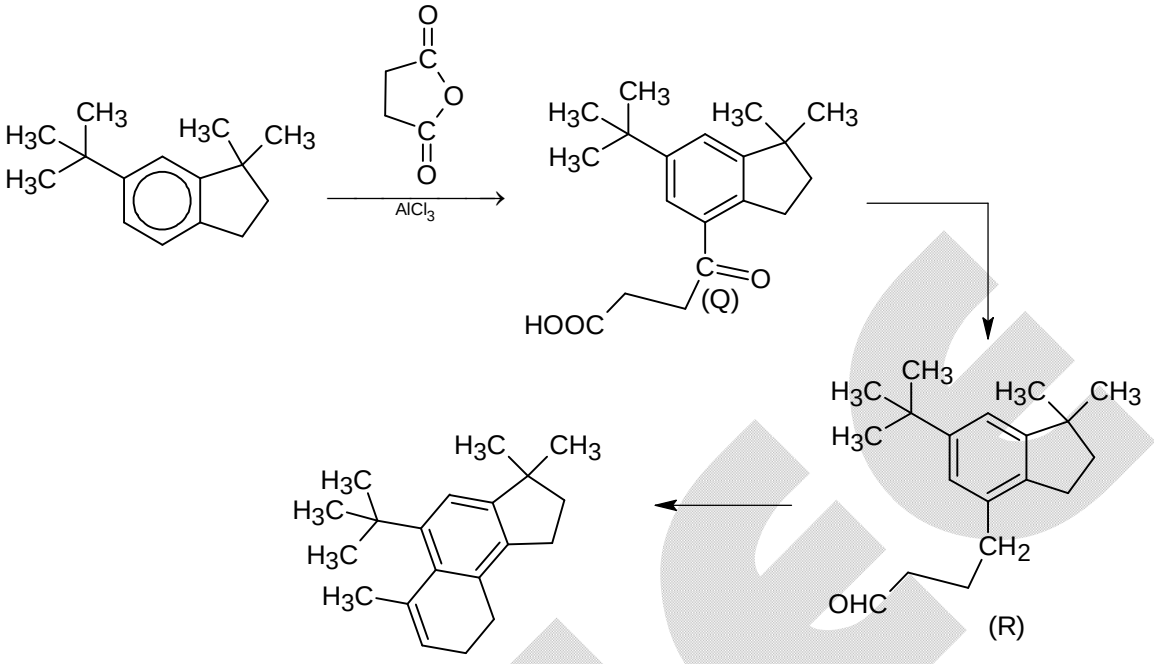
The total surface area $= 6\times(1\text{ mm})^2\times10^9 = 6\times10^9$ mm^2

$\therefore$ Increase in surface area $= \frac{6\times10^9}{6\times10^6} = 10^3$ mm^2

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58. 7

Sol.



59. 6

Sol.

$$FeSO_4 + 6KCN(\text{excess}) \longrightarrow K_4\left[Fe(CN)_6\right]SO_4 + K_2SO_4$$

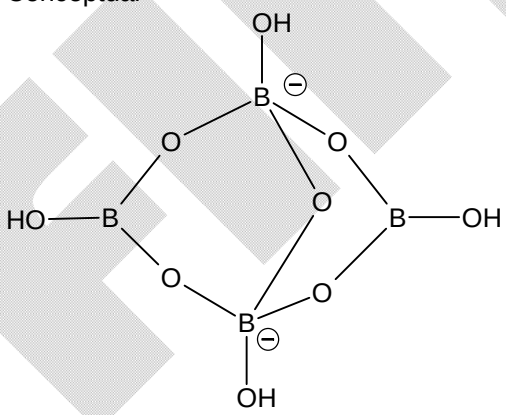
$$K_4\left[Fe(CN)_6\right] + 6H_2SO_4\,(\text{conc.}) + 6H_2O \longrightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2\,SO_4 + 6CO$$

$$3K_4\left[Fe(CN)_6\right] + 6H_2SO_4\,(\text{dilute}) \longrightarrow 12HCN + Fe_2\left[Fe(CN)_6\right] + 6K_2SO_4$$

60. 13

Sol. $\begin{cases} p = 2,\ x = 3,\ y = 2,\ z = 6 \\ Be_3Al_2\,(SiO_3)_6 \end{cases}$

Conceptual



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Mathematics

PART – C

SECTION – A

61. B

Sol. $(\sin x + \cos x)\left(\frac{1}{2}+\cos x\right)=0 \Rightarrow$ 4 real solutions.

62. C

Sol. $5c = f^3$ and $3c = g^2$

63. B

Sol. $g(x) = x^7 + x^3$

64. D

Sol. P lies on the directrix

$\Rightarrow h = -2$

65. D

Sol. $\sin^{-1}|\cos x| = \cos^{-1}|\sin x| \quad \forall\, x \in R.$

66. C

Sol. $\int_0^1 \frac{x^a - x^b}{\log x}dx = \int_0^1 \frac{x^a - 1}{\log x}dx - \int_0^1 \frac{x^b - 1}{\log x}dx = \log\left(\frac{1+a}{1+b}\right)$

67. D

Sol. $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma\left(\vec{a}\times\vec{b}\right)$

$\alpha = \beta = \cos\theta$

$\vec{c}\cdot\vec{c} = 2\cos^2\theta + \gamma^2 \Rightarrow \gamma^2 = -\cos 2\theta$

68. D

Sol. $N = 441000 = 2^3.3^2.5^2.7^2$

69. B

Sol. $P \equiv (1, 2, 3)$ and $Q \equiv (2, 1, 2)$

70. A

Sol. $\frac{x\,dx + y\,dy}{\cos^2(x^2+y^2)} = \frac{x}{y}\left(\frac{y\,dx - x\,dy}{y^2}\right)$

$\Rightarrow \tan\left(x^2+y^2\right) = \left(\frac{x}{y}\right)^2 + c$

71. C

Sol. Let $A = B + I$

$B = \begin{bmatrix}0&0&0\\1&0&0\\1&1&0\end{bmatrix} \quad B^2 = \begin{bmatrix}0&0&0\\0&0&0\\1&0&0\end{bmatrix} \quad B^3 = \begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$

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$$A^{20} = I + 20B + 190B^2 = \begin{bmatrix} 1 & 0 & 0 \\ 20 & 1 & 0 \\ 210 & 20 & 1 \end{bmatrix}$$

72. C

Sol. S = we are creating a team containing n players with a position of captain and wicket keeper out of n boys and n girls where boy will lead that team and same person can hold both the positions.

73. B

Sol. Circumcentre of ΔABC is (0, 0)

$\Rightarrow (\alpha, \beta) \equiv \left(2\sqrt{5} + \sqrt{13} - 3, \sqrt{5} + 2\sqrt{3} + 4\right)$

$\Rightarrow [\alpha] = 5$

74. C

Sol. $\sqrt{k}$ must be irrational for exactly one solution

$\Rightarrow$ k = 2, 3, 5, 6, 7, 8, 10

75. C

Sol. Let M be the foot of perpendicular from point Q to the directrix

PQ + QS = PQ + QM $\geq$ PM.

76. C

Sol. Let A = sin1°sin3° ……. sin89°

B = sin2°sin4°sin6°……. sin88°

$\Rightarrow$ B = 2^{44} sin1°cos1°sin2°cos2°……sin44°cos44°

$\Rightarrow$ B = 2^{44}.AB.$\sqrt{2}$

$\Rightarrow$ A = $2^{-\frac{89}{2}}$

77. C

Sol. Statement is always true or always false.

78. B

Sol. $|\vec{a} - \vec{c}|^2 = |\vec{b} - \vec{c}|^2$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \cdot \left(\vec{a} - \vec{b}\right) = 2\vec{c} \cdot \left(\vec{a} - \vec{b}\right)$

$\Rightarrow \left(\vec{a} - \vec{b}\right) \cdot \left[\frac{\left(\vec{a} + \vec{b}\right)}{2} - \vec{c}\right] = 0$

79. B

Sol. $x - 5^{\log_{10} x} = 2^{\log_{10} x}$

Let $\log_{10} x = t$

$\Rightarrow 10^t - 5^t = 2^t$

$\Rightarrow 2^t + 5^t = 10^t$

$\Rightarrow$ only one real solution.

80. A

Sol. $\int \frac{x^6 + 4x^3}{\left(x^4 + x^3 + 1\right)^2} dx = -\frac{1 + x^3}{x^4 + x^3 + 1} + c$

$\Rightarrow$ f''(–1) = 6, λ = 0

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SECTION – B

81. 0

Sol. Let $2x^3 - 9x^2 + 12x + 1 = t$

$\Rightarrow t^2 - 2kt + k^2 - 1 = 0$

For 6 real roots, both roots of the above quadratic in 't' must lie between 5 and 6.

$\Rightarrow k \in \phi$.

82. 14

Sol. $$\sum_{k=1}^{n} \frac{2^k\left(k^2+2k+2\right)}{(k+1)(k+2)} = \sum_{k=1}^{n}\left(\frac{(k+1)2^{k+1}}{k+2} - \frac{k\cdot 2^k}{k+1}\right) = 2^{n+1} - \frac{2^{n+1}}{n+2} - 1$$

83. 6

Sol. The line must pass through centre of the two degree curve

$\Rightarrow \lambda = -6$

84. 16

Sol. $k = 6 + 13I$ or $k = 7 + 13I$, $I \in$ integer.

85. 40

Sol. $a_0 = 2^{20}$, $a_1 = 20 \times 2^{20}$, $a_{39} = 40$

86. 2

Sol. $$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}\sin x}{\sqrt{1-\cos^3 x}}dx$$

Let $\cos^{3/2}x = t$

$$\Rightarrow \frac{3}{2}\cos^{1/2}x.\sin x\, dx = -dt$$

$$\Rightarrow I = -\int_1^0 \frac{2}{3}\frac{dt}{\sqrt{1-t^2}} = \frac{2}{3}\int_0^1 \frac{dt}{\sqrt{1-t^2}}$$

$$\Rightarrow I = \frac{2}{3}\sin^{-1}(t)\Big|_0^1 = \frac{\pi}{3}$$

87. 1

Sol. $$\frac{k_1(8000)+k_2(9000)}{k_1+k_2} = 8400$$

$$\Rightarrow 600k_2 = 400k_1 \Rightarrow \frac{k_1}{k_2} = \frac{3}{2}$$

88. 31

Sol. $x^2 - 10x + 16 > 0$

$\Rightarrow x < 2$ and $x > 8$

$\Rightarrow$ required probability $= \frac{93}{100}$

89. 1

Sol. Only one such triangle is possible for $b = \frac{9}{5}$

90. 0

Sol. $\alpha^2 + \alpha + 1 < 0$ and $\alpha^2 - \alpha - 1 < 0$ and $\alpha^4 - (\alpha + 1)^2 - 2 < 0$

$\Rightarrow \alpha \in \phi$.

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